Shubhadeep Mandal

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Education

International Institute of Information Technology, Hyderabad July 2025 – Present
Telangana, India
Master of Technology in Computer Science and Engineering *CGPA: 9.12*

Indian Institute of Information Technology, Ranchi Aug 2020 – Jun 2024
Ranchi, India
Bachelor of Technology (Hons.) in Computer Science and Engineering *CGPA: 9.05*

Projects

Warehouse Traffic Control: 3D Distributed Systems Simulator | *JavaScript, Three.js, Web Workers, Vite*

- Built a live 3D browser simulator of a 200-robot warehouse in Three.js, deployed to GitHub Pages via GitHub Actions, where robots and managers interact only over a lossy, delayed network.
- Designed a lease-and-fencing protocol across 16 region managers, sustaining **0 collisions over 416K moves** under 40% message loss, ±20% clock drift and injected crashes.
- Built checkpoint-based replay that rewinds to any moment in **under 0.2 s**, and ran 56 benchmark simulations in parallel on Web Workers.

Negotiate, Then Solve: Full-Stack Scheduling Platform | *React, TypeScript, FastAPI, Python, SQLite*

- Built a React 19 + TypeScript front end over a FastAPI JSON API with server-enforced role-based permissions, negotiation inboxes, an approval queue, audit traces and version rollback.
- Backed the app with **215 pytest tests** and persistence across restarts; a solver cache cut demo replay from **2.5 min to 25 s**.
- Integrated an LLM agent pipeline (Gemini, RAG) that turns free-text requests from email, chat and the web portal into typed constraints, with guardrails under which **no attack succeeded** across 50 prompt-injection and spoofing tests.

BoardBuddy: Real-Time Collaborative Whiteboard | *Next.js 14, Convex DB, Liveblocks, TypeScript*

- Developed a full-stack collaborative canvas supporting multi-user synchronization, freehand sketching, geometric shapes, and undo/redo.
- Synchronized real-time state and user presence across sessions using Liveblocks WebSockets and Convex reactive database.
- Secured organization-scoped workspaces and role permissions via Clerk authentication and responsive TailwindCSS/shadcn UI.

Peer-to-Peer Distributed File Sharing System | *C++, POSIX Sockets, Multithreading, OpenSSL, Linux*

- Engineered a BitTorrent-style hybrid P2P system in C++, achieving an **8.5x transfer speedup on 1GB files** (193s to 22.8s) across 5 concurrent peers by saturating network and disk I/O.
- Implemented out-of-order parallel chunk writing directly to designated file offsets and dynamic bitfield generation for real-time partial seeding during active downloads.
- Architected Primary-Backup tracker failover with 2s bounded failover, full state synchronization, and verified resilience via **48 unit tests** under AddressSanitizer/UBSan.

Technical Skills

Languages: TypeScript, JavaScript, Python, C/C++, SQL, HTML/CSS, Bash.

Frontend Engineering: React, Next.js 14, TailwindCSS, shadcn/ui, Three.js, Web Workers, Vite, Canvas API.

Backend & Data: FastAPI, Node.js, REST APIs, Convex, Prisma ORM, PostgreSQL, SQLite, Clerk Auth.

Real-Time, Testing & Tools: WebSockets (Liveblocks), pytest, Git, GitHub Actions CI/CD, Docker, Vercel, Linux.

Achievements

- **B.Tech Rank Holder:** Ranked 2nd in IIIT Ranchi by GPA in B.Tech CSE.
- **PGEE 2025:** Secured All-India Rank 100 in IIIT-Hyderabad Post Graduate Entrance Examination (PGEE).
- **Megathon 2025:** Third rank team in Megathon 2025 in the mobile computing track.

Interests

- Web Novel and Manga Enthusiast.
- Chess and Badminton.